

Exercises: Differentiation and Integration

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Question 1 — Differentiation

For each of the following functions, calculate $f'(x)$.

a) $f(x) = \frac{5}{1-x}$

We will use the quotient rule:

$$\left(\frac{u}{v}\right)' = \frac{u'v - u v'}{v^2}$$

$$\begin{aligned}u(x) &= 5, \\v(x) &= 1 - x, \\u'(x) &= 0, \\v'(x) &= -1\end{aligned}$$

$$\begin{aligned}\left(\frac{u}{v}\right)' &= \frac{0(1-x) - 5(-1)}{(1-x)^2} \\&= \frac{0+5}{(1-x)^2} \\&= \frac{5}{(1-x)^2}\end{aligned}$$

b) $f(x) = \frac{1-x}{5}$

We will use the quotient rule:

$$\left(\frac{u}{v}\right)' = \frac{u'v - u v'}{v^2}$$

$$\begin{aligned}u(x) &= 1 - x, \\v(x) &= 5, \\u'(x) &= -1, \\v'(x) &= 0\end{aligned}$$

$$\begin{aligned}
\left(\frac{u}{v}\right)' &= \frac{(-1)(5) - (1-x)(0)}{(5)^2} \\
&= \frac{-5}{25} \\
&= -\frac{1}{5}
\end{aligned}$$

c) $f(x) = \sin 2x + \cos 3x$

The derivative of a sum is the sum of the derivatives.

$$u(x) = \sin 2x,$$

$$v(x) = \cos 3x,$$

For each function we will use the chain rule:

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

First term, with $u(x) = \sin 2x$:

$$\begin{aligned}
u'(x) &= \cos 2x \cdot 2 \\
&= 2 \cos 2x
\end{aligned}$$

Second term, with $v(x) = \cos 3x$:

$$\begin{aligned}
v'(x) &= -\sin 3x \cdot 3 \\
&= -3 \sin 3x
\end{aligned}$$

Putting this together:

$$\begin{aligned}
f'(x) &= u'(x) + v'(x) \\
&= 2 \cos 2x - 3 \sin 3x
\end{aligned}$$

d) $f(x) = \sin\left(4x + \frac{\pi}{2}\right)$

Here we will use the chain rule:

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

$$u(x) = \sin(\cdot),$$

$$v(x) = 4x + \frac{\pi}{2},$$

Applying the chain rule:

$$\begin{aligned}f'(x) &= \cos\left(4x + \frac{\pi}{2}\right) \cdot 4 \\&= 4 \cos\left(4x + \frac{\pi}{2}\right)\end{aligned}$$

Alternatively, we can use the identity $\sin\left(\theta + \frac{\pi}{2}\right) = \cos \theta$ with $\theta = 4x$, we can simplify:

$$f(x) = \sin\left(4x + \frac{\pi}{2}\right) = \cos 4x$$

Then use the chain rule to differentiate:

$$\begin{aligned}f'(x) &= -\sin 4x \cdot 4 \\&= -4 \sin 4x\end{aligned}$$

e) $f(x) = \sin 2x \cos 3x$

Here we will use the product rule overall:

$$(uv)' = u'v + uv'$$

and the chain rule for each function.

$$\begin{aligned}u(x) &= \sin 2x, \\v(x) &= \cos 3x,\end{aligned}$$

For each function we will use the chain rule:

$$(g(h(x)))' = g'(h(x)) \cdot h'(x)$$

First term, with $u(x) = \sin 2x$:

$$\begin{aligned}u'(x) &= \cos 2x \cdot 2 \\&= 2 \cos 2x\end{aligned}$$

Second term, with $v(x) = \cos 3x$:

$$\begin{aligned}v'(x) &= -\sin 3x \cdot 3 \\&= -3 \sin 3x\end{aligned}$$

Putting this together:

$$\begin{aligned}f'(x) &= u'(x)v(x) + u(x)v'(x) \\&= (2 \cos 2x) \cdot (\cos 3x) + (\sin 2x) \cdot (-3 \sin 3x) \\&= 2 \cos 2x \cos 3x - 3 \sin 2x \sin 3x\end{aligned}$$

f) $f(x) = \sin(2 \cos 3x)$

$$u(x) = \sin(\cdot),$$

$$v(x) = 2 \cos 3x,$$

We will use the chain rule:

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

Applying the chain rule:

$$\begin{aligned} f'(x) &= u'(v(x)) \cdot v'(x) \\ &= \cos(2 \cos 3x) \cdot (2)(-\sin 3x)(3) \\ &= -6 \cos(2 \cos 3x)(\sin 3x) \end{aligned}$$

Question 2 — Integration

For each of the following functions, calculate $\int f(x) \, dx$.

a) $f(x) = x^4 - x^3 + x^2$

This is a polynomial, so we apply the power rule for antiderivatives:

Theorem 1. If $f(x) = \sum_{i=0}^n c_i x^i$, then $\int f(x) \, dx = \sum_{i=0}^n \frac{1}{i+1} c_i x^{i+1} + c$.

This holds whenever every exponent i is a non-negative integer, so that $i+1 \neq 0$. Here the exponents are 2, 3, 4, so this rule applies.

Note: that this theorem combines two facts: the power rule applied to each individual term, and the linearity of integration, which lets us split the sum and factor out each coefficient c_i . We apply it term by term below.

$$\begin{aligned} \int f(x) \, dx &= \int (x^4 - x^3 + x^2) \, dx \\ &= \frac{1}{5}x^5 - \frac{1}{4}x^4 + \frac{1}{3}x^3 + c \end{aligned}$$

b) $f(x) = \frac{3}{x}$

Since the integrand has x to the power -1 , the power rule from 2a) cannot be applied here as it would require dividing by $i+1=0$. Here we need to use the logarithm rule:

$$\int \frac{1}{x} \, dx = \ln |x| + c.$$

Pulling out any constants, e.g:

$$\int \frac{a}{x} \, dx = a \int \frac{1}{x} \, dx$$

Putting this all together:

$$\begin{aligned}\int f(x) \, dx &= \int \frac{3}{x} \, dx \\ &= 3 \int \frac{1}{x} \, dx \\ &= 3 \ln |x| + c\end{aligned}$$

c) $f(x) = 2 \sin x + 3 \cos x$

Here we can rely on the known standard derivatives of $\sin$ and $\cos$ to work out their antiderivatives. Since differentiation is linear, the derivative of a sum is the sum of the derivatives, and constants factor out. The same holds for antiderivatives, so we can integrate term by term.

$$\begin{aligned}\int f(x) \, dx &= \int (2 \sin x + 3 \cos x) \, dx \\ &= 2 \int \sin x \, dx + 3 \int \cos x \, dx \\ &= 2(-\cos x) + 3(\sin x) + c \\ &= -2 \cos x + 3 \sin x + c\end{aligned}$$

d) $f(x) = 5e^x - e$

Here we can rely on the known standard derivative of e^x to work out these antiderivatives. Since differentiation is linear, the derivative of a sum is the sum of the derivatives, and constants factor out. The same holds for antiderivatives, so we can integrate term by term.

$$\begin{aligned}\int f(x) \, dx &= \int (5e^x - e) \, dx \\ &= 5 \int e^x \, dx - \int e \, dx \\ &= 5(e^x) - (ex) + c \\ &= 5e^x - ex + c\end{aligned}$$

e) $f(x) = 2x \sin 4x$

In this exercise we have a product of 2 functions $2x$ and $\sin 4x$ so we cannot rely on any of the past rules from examples $a - d$. Here we need to use the formula for integration by parts:

$$\int g(x) h'(x) \, dx = g(x) h(x) - \int g'(x) h(x) \, dx$$

$$\begin{aligned}g(x) &= 2x \\ h'(x) &= \sin 4x \\ h(x) &= -\frac{1}{4} \cos 4x \\ g'(x) &= 2\end{aligned}$$

$$\begin{aligned}
\int f(x) \, dx &= g(x) h(x) - \int g'(x) h(x) \, dx \\
&= (2x) \left(-\frac{1}{4} \cos 4x\right) - \int 2 \left(-\frac{1}{4} \cos 4x\right) \, dx \\
&= -\frac{1}{2} x \cos 4x + \frac{1}{2} \int \cos 4x \, dx \\
&= -\frac{1}{2} x \cos 4x + \left(\frac{1}{2}\right) \left(\frac{1}{4}\right) \sin 4x + c \\
&= -\frac{1}{2} x \cos 4x + \frac{1}{8} \sin 4x + c
\end{aligned}$$

f) $f(x) = x^2 e^x$

In this exercise we have a product of 2 functions x^2 and e^x so we cannot rely on any of the past rules from examples $a - d$. Here we need to use the formula for integration by parts:

$$\int g(x) h'(x) \, dx = g(x) h(x) - \int g'(x) h(x) \, dx$$

$$\begin{aligned}
g(x) &= x^2 \\
h'(x) &= e^x \\
h(x) &= e^x \\
g'(x) &= 2x
\end{aligned}$$

$$\begin{aligned}
\int f(x) \, dx &= g(x) h(x) - \int g'(x) h(x) \, dx \\
&= x^2 e^x - \int 2x e^x \, dx
\end{aligned}$$

Now we need to apply the integration by parts rule again but for $u(x) = 2x$ and $v(x) = e^x$.

Let $z(x) = 2x e^x$

$$\begin{aligned}
u(x) &= 2x \\
v'(x) &= e^x \\
v(x) &= e^x \\
u'(x) &= 2
\end{aligned}$$

$$\begin{aligned}
\int z(x) \, dx &= u(x) v(x) - \int u'(x) v(x) \, dx \\
&= (2x)(e^x) - \int (2)(e^x) \, dx \\
&= 2x e^x - 2 \int e^x \, dx \\
&= 2x e^x - 2e^x + c
\end{aligned}$$

Now we can substitute this into our original integration by parts for $f(x)$:

$$\begin{aligned}\int f(x) \, dx &= g(x) h(x) - \int g'(x) h(x) \, dx \\ &= x^2 e^x - \int 2x e^x \, dx \\ &= x^2 e^x - (2xe^x - 2e^x) + c \\ &= x^2 e^x - 2xe^x + 2e^x + c\end{aligned}$$